

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA02

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Class Test

Code of Conduct:

I CH.V.Ramayogi(192424319)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor	Partially correct with missing	Incorrect answers with major gaps	

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Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	

Annexure A

Class Test

Question 1: Image Preprocessing and Feature Extraction

Scenario: A drone is used to capture aerial images of a farmland to identify pest affected crop regions. Due to sensor limitations, wind movement, and changing sunlight, the captured images contain noise, shadows, and uneven illumination. Pest affected areas usually show visible texture and color variations compared to healthy crop regions.

Answer the following questions:

a) Preprocessing Pipeline — 4 Marks

Propose a suitable preprocessing pipeline to improve the quality of the farmland images before analysis. Your answer should include methods for noise reduction, illumination correction, contrast enhancement, and normalization. Justify the purpose of each step.

b) Feature Extraction — 3 Marks

Choose an appropriate feature extraction method to identify texture changes caused by pest damage. Explain why the selected method is suitable for detecting affected crop regions.

c) Rotation and Scale Variations — 3 Marks

Drone images may be captured from different heights and angles. Discuss how rotation and scale variations can affect feature detection. Suggest suitable methods to make the feature extraction process more robust.

Question 2: Stereo Vision and Motion Estimation

Scenario: A self-driving car is equipped with stereo cameras to navigate through a busy street. The system must estimate the distance of nearby vehicles and pedestrians in real time. It should also track moving objects accurately to avoid collisions, especially when pedestrians or vehicles change speed suddenly.

Answer the following questions:

a) Stereo Disparity and Depth Estimation — 4 Marks

Explain how stereo cameras use the difference between left and right image views to estimate the depth of objects. Discuss how disparity changes for nearby and distant objects.

b) Optical Flow for Motion Tracking — 3 Marks

If a pedestrian suddenly changes speed or direction, explain how optical flow helps in tracking the pedestrian's motion between consecutive video frames. c) Errors and Mitigation

Techniques — 3 Marks

Identify possible sources of error in depth estimation and motion tracking, such as camera misalignment, poor lighting, occlusion, fast motion, or textureless regions. Suggest suitable techniques to reduce these errors.

ANSWERS:

Question 1 (a) Preprocessing Pipeline (4 Marks)

Answer:

Drone images captured for pest detection often suffer from noise, uneven illumination, and low contrast, which reduce the accuracy of feature extraction and classification. Hence, a preprocessing pipeline is applied before analysis.

Step 1: Noise Removal

A Median Filter or Gaussian Filter is applied to remove sensor noise and small artifacts introduced by the drone camera or atmospheric conditions. This smooths the image while preserving important edges.

Step 2: Illumination Correction

Since images are captured under varying sunlight conditions, Histogram Equalization or CLAHE (Contrast Limited Adaptive Histogram Equalization) is used to normalize brightness across the image, correcting shadows and overexposed regions.

Step 3: Contrast Enhancement

CLAHE is applied locally to enhance contrast between healthy and pest-affected regions, making texture and color differences more distinguishable.

Step 4: Normalization

Pixel intensity values are scaled to a common range (e.g., 0–1 or 0–255) so that all images have consistent intensity distribution, ensuring feature extraction algorithms perform reliably across the dataset.

Conclusion: This four-stage pipeline (denoising → illumination correction → contrast enhancement → normalization) improves image quality and significantly increases the reliability of downstream pest detection.

Question 1 (b) Feature Extraction (3 Marks)

Answer:

Local Binary Pattern (LBP) is used to extract texture features for distinguishing healthy crops from

pest-affected ones.

Working principle: For each pixel, LBP compares its intensity with its surrounding neighbors, encoding the result as a binary pattern which is converted into a decimal value representing the local texture.

Why LBP is suitable here:

- Healthy crop regions exhibit uniform, repetitive texture, while pest-damaged regions show irregular, rough texture patterns.
- LBP is computationally efficient and suitable for real-time drone-based analysis.
- It is robust to monotonic illumination changes, which is important given varying outdoor lighting.

Conclusion: LBP effectively differentiates healthy and damaged crop textures, making it an appropriate feature extraction technique for this application.]

Question 1 (c) Rotation and Scale Variations (3 Marks)

Answer:

Drone altitude and camera angle vary during flight, causing the same crop region to appear at different scales and orientations across images. This variation makes simple pixel-based comparison unreliable.

Solution: Scale- and rotation-invariant feature descriptors are used:

- SIFT (Scale-Invariant Feature Transform): Detects keypoints that remain stable across scale and rotation changes.
- SURF (Speeded-Up Robust Features): A faster alternative to SIFT with similar invariance properties.
- ORB (Oriented FAST and Rotated BRIEF): A computationally lightweight, rotation-aware descriptor suited for real-time drone applications.

Conclusion: These descriptors allow the same crop feature to be correctly matched regardless of drone altitude or viewing angle, improving detection accuracy and consistency.

Question 2 (a) Stereo Disparity and Depth Estimation (4 Marks)

Answer:

A stereo vision system uses two cameras positioned at a fixed horizontal baseline, capturing the same scene from slightly different viewpoints (left and right).

Disparity: The positional difference of a corresponding object point between the left and right images.

Depth relationship:

$$\text{Depth} \propto \frac{1}{\text{Disparity}} \quad \text{Depth} \propto \frac{1}{\text{Disparity}}$$

- Nearby objects → large pixel shift between the two views → large disparity
- Distant objects → small pixel shift → small disparity

Using the known camera baseline and focal length, the system computes disparity and converts it into an accurate depth (distance) estimate for each object in the scene.

Application: This allows a self-driving car to determine how far away vehicles, pedestrians, and obstacles are, enabling safe navigation and collision avoidance.

Question 2 (b) Optical Flow for Motion Tracking (3 Marks)

Answer:

Optical Flow estimates the apparent motion of pixels between two consecutive video frames, based on changes in pixel intensity over time.

Working: When a pedestrian changes speed or direction, the pixels corresponding to that pedestrian shift position between frames. Optical flow algorithms (e.g., Lucas-Kanade, Farneback) compute a motion vector for these pixels, indicating both speed and direction of movement.

Application: By continuously tracking these motion vectors, the self-driving car can predict the pedestrian's next likely position and react in advance, improving collision avoidance and overall road safety.

Question 2 (c) Errors and Mitigation Techniques (3 Marks)

Answer:

There are several factors that can reduce the accuracy of depth estimation and motion tracking.

If the cameras are not properly aligned, the calculated depth may be incorrect. This problem can be solved by camera calibration.

Poor lighting or strong shadows can make object detection difficult. We can reduce this problem using Histogram Equalization or CLAHE.

Sometimes one object blocks another object, which is called occlusion. In this case, tracking methods such as the Kalman Filter can predict the object's movement.

Fast-moving objects may appear blurred, so using high-frame-rate cameras and Optical Flow improves tracking accuracy.

In textureless regions where there are very few image details, methods such as SIFT or ORB help detect

reliable features.

By applying these techniques, we can improve the accuracy of both depth estimation and motion tracking.